

CSES — Construction Problems Reference

All tasks: 1 s, 512 MiB. Any valid construction is accepted unless stated otherwise.

2214 — Inverse Inversions

Construct a permutation p of $1, \dots, n$ having exactly k inversions, where an inversion is a pair $i < j$ with $p_i > p_j$.

In: n k

Out: one valid permutation

Constraints: $1 \leq n \leq 10^6$, $0 \leq k \leq n(n-1)/2$.

2215 — Monotone Subsequences

For each test, construct a permutation of $1, \dots, n$ whose *longest monotone subsequence* has exactly k elements. A monotone subsequence may be increasing or decreasing.

In: t ; then n k

Out: a valid permutation for each test, or IMPOSSIBLE

Constraints: $t \leq 1000$, $1 \leq k \leq n \leq 100$.

3422 — Third Permutation

Given permutations a, b with $a_i \neq b_i$ for every position, construct a permutation c such that simultaneously $c_i \neq a_i$ and $c_i \neq b_i$ for every i .

In: n ; permutation a ; permutation b

Out: a valid permutation c , or IMPOSSIBLE

Constraints: $2 \leq n \leq 10^5$.

3423 — Permutation Prime Sums

Construct two permutations a, b of $1, \dots, n$ such that $a_i + b_i$ is prime for every position i .

In: n

Out: the two permutations, or IMPOSSIBLE

Constraints: $1 \leq n \leq 10^5$.

1697 — Chess Tournament

Player i wants to play exactly x_i games. Choose unordered pairs of distinct players so every player gets the requested degree; each pair may play at most once.

In: n ; $x_1 \dots x_n$

Out: number k of games and the k player pairs; or IMPOSSIBLE

Constraints: $n \leq 10^5$, $\sum_i x_i \leq 2 \cdot 10^5$.

3424 — Distinct Sums Grid

Construct an $n \times n$ integer grid such that:

- every value $1, \dots, n$ appears exactly n times overall;
- the n row sums and n column sums are all pairwise distinct, giving exactly $2n$ distinct sums.

In: n

Out: a valid grid, or IMPOSSIBLE

Constraints: $1 \leq n \leq 1000$.

2423 — Filling Trominos

Tile the entire $n \times m$ grid with L-trominos, each occupying exactly three cells in an L shape. If possible, output letters so two edge-adjacent cells have the same letter *iff* they belong to the same tromino.

In: t ; then n m

Out: for each test, YES plus an n -row letter grid, or NO

Constraints: $t \leq 100$, $1 \leq n, m \leq 100$.

2418 — Grid Path Construction

Given distinct squares $a = (y_1, x_1)$ and $b = (y_2, x_2)$ in an $n \times m$ grid, construct a path from a to b that visits *every grid square exactly once*. Moves are only U/D/L/R.

In: t ; then n m y_1 x_1 y_2 x_2

Out: for each test, YES and a move string of length $nm - 1$, or NO

Constraints: $t \leq 100$, $1 \leq n, m \leq 50$; endpoints are distinct.